

Material Properties

Peroxide dynamic crosslinking in impact modification of polypropylene with polybutadiene



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ABSTRACT

Polypropylene was blended with polybutadiene in an internal mixer in order to improve its toughness. The rubber content ranged from 10 to 20 wt. % and dicumyl peroxide (DCP) was added to provide required radicals for dynamic crosslinking of polybutadiene, and for in-situ compatibilization of the phases. This was done using two sequences of mixing. In addition, zinc dimethacrylate was utilized as a co-agent to control chain scission of PP, enhance the interfacial reactions and increase compatibility of the components. In obtaining the optimum blend, the Taguchi orthogonal experimental design was applied and results of Izod impact strength test were concordantly related to crystallinity of the polypropylene, morphology of the blends and reactions that may occur. The impact strength was increased about four times at best in comparison to the neat polypropylene, showing versatility of this way of toughening which may be improved further.

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1. Introduction

Polypropylene (PP) is a semi-crystalline polymer with outstanding mechanical, thermal and chemical properties that makes it a widely-used material [1]. However, its low toughness, particularly at low temperatures, has limited the structural applications of this commodity polymer. Low impact strength of PP is attributed to its crystalline structure [2] and several strategies have been devised to overcome this shortcoming. These include: (i) filling PP with rigid organic particles (ROP), (ii) filling PP with rigid inorganic particles (RIP), (iii) blending PP with rubbers, (iv) filling PP–rubber blends with ROP or RIP, and (v) random copolymerization of PP [3,4]. Among these methods, using rubber as an impact modifier is a very common approach to improve the impact resistance of PP. It is well-known that mechanical performance of polymer blends, which are usually immiscible, depends to a large extent to the compatibility of their components. This is why EPDM is the most often studied elastomer for increasing PP impact strength [5–14]. Nonetheless, increasing polypropylene/elastomer compatibility is still under investigation. In this respect, using a peroxide curing system and co-agents has been found effective.

Chen et al. [10] reported a positive role of zinc dimethacrylate (ZDMA) on EPDM toughened PP that was dynamically crosslinked with DCP. It reduced size of the rubbery phase and enhanced interfacial interactions, which was reflected in mechanical as well as rheological properties. Cao et al. [15] studied in situ reactive compatibilization of TPVs based on PP/EPDM using ZDMA. They showed that ZDMA reinforces EPDM particles. Substantial improvements in the rheological and mechanical properties of the resultant materials imply increasing compatibility. The use of peroxide/cogent systems has been extended to other elastomers. It was shown that ZDMA reduces PP degradation due to DCP and improves the mechanical properties of blends [15–18]. Xu et al. tried to toughen polypropylene using nitrile butadiene rubber (NBR) and ZDMA in the presence of DCP [18]. The drop and increment of mixing curves after the addition of DCP and ZDMA, respectively, was assigned to the PP degradation and ZDMA polymerization. In addition, they showed the possibility of nucleation for PP crystallization owing to addition of ZDMA. To increase impact resistance of PP with SBS, Yang et al. [19] utilized a reactive process in combination with dicumyl peroxide (DCP) as an initiator and cross linker of SBS, and triallyl isocyanurate (TAIC) as a compatibilizer and co-crosslinker to prevent PP chain scission. Dispersed SBS particles sizes were reduced and their adhesion to the PP matrix amplified, resulting in better mechanical properties. Xu, Cao et al. [20] developed an approach to compatibilize

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polypropylene (PP) and acrylonitrile butadiene rubber (NBR) using maleic anhydride grafted polypropylene (MA-g-PP) as a compatibilizer in the presence of zinc dimethacrylate (ZDMA). Interfacial adhesion of PP/NBR increased due to of ZDMA polymerization in combination with MA-g-PP, which results in considerable improvement in mechanical properties.

Apart from employing co-agents, various strategies have been utilized to enhance compatibility in polypropylene blends. Liu and Qiu [21] added polyolefin elastomer (POE) to increase toughness of PP and found it as an effective method for impact modification. However, POE efficiency increases using a small amount of HDPE. Zebarjad et al. [22] focused on the role of EPR particles on the impact properties of PP/EPR blends. The results showed that impact strength rises with both EPR content and test temperature, showing a brittle-to-ductile transition temperature (BDTT). Also, they illustrated that the dominant fracture mechanism is the formation of craze-like structures that are highly localized dilatational bands. Maleated EPDM (EPDM-g-MAH) and maleated PP (PP-g-MAH) were used in various concentrations as compatibilizer [5]. Kim et al. [6] studied the effect of (PP-g-MAH)-co-[PA-6, 6]-co-(EPDM-g-MAH) on impact properties of PP/EPDM blends and related it to formation of β -form crystals.

On the other hand, poly (*cis*-butadiene) rubber (polybutadiene/PB), which is a non-polar polymer, exhibits excellent elastic properties even at low temperatures. However, it is rarely utilized as a toughener for PP, possibly due to its inherent immiscibility. In this case, the phase structure and morphology of binary blends of isotactic polypropylene (iPP) with PB and partly compatible iPP with ethyl acrylate grafting onto poly (*cis*-butadiene) rubber (PB-g-EA) were studied by Sheng, Ma and coworkers [23–26]. In addition, Jalali et al. [27] studied effect of blend ratios, curing system amount and advantages of dynamic vulcanization on overall properties of PP/PB TPVs. In these studies, PB without compatibilizer results in minor increase in compatibility, and thus PP impact strength.

In this research, potential of PB in increasing the impact strength of PP is examined. The compatibility was investigated in terms of mechanical, morphological and physical properties. The main goal is to find an alternative to EPDM, for impact modification of PP. In this respect, PB was chosen because of both its availability and lower price.

2. Experimental

Materials used in this study were polypropylene, HP550J, from Jam Polypropylene Co. (MFR at 230 °C, 2.16 kg = 3.2 g/10 min, density = 0.91 g/cm³), polybutadiene, PBR-1220, from Shazand Petrochemical Co. (97% *cis*, ML4 = 45), zinc dimethacrylate from

Sigma Aldrich and dicumyl peroxide, PEROXAN DC-P, from Pergan Co. All chemicals were used as received.

Mixing of PP and PB was done in an internal mixer, Brabender W 50 EHT, using two procedures designated sequence of mixing 1 and 2. In sequence 1, firstly, ZDMA was added to PP/PB during their blending, and afterwards, DCP was added to the compound. In sequence 2, firstly, DCP was dispersed in PB at room temperature in the internal mixer, and then ZDMA was added during melt mixing of PP and DCP containing PB. The mixing temperature was 170 °C and the rotor speed was 60 rpm.

Subsequently, 4 mm width sheets were compression molded from the above blends after cooling. Eventually, test specimens were prepared from these sheets.

The Taguchi orthogonal experimental design was exploited to lessen the number of experiments using the Taguchi L₁₈ orthogonal array. Chosen factors were sequence of mixing, and the amounts of rubber, peroxide agent and ZDMA. Table 1 summarizes the experimental design as well as the reference samples. The best level of each factor was determined according to average of the impact strength results.

Izod impact strength was measured at room temperature according to ISO 180 by a Ray-Ran impact tester. For all of the samples, impact strength was determined by an average of the three specimens' results.

Differential scanning calorimeter was utilized to investigate PP crystallinity. This was carried out using a Mettler Toledo DSC 822^e. All measurements were performed under a nitrogen purge to minimize degradation. In each run, a sample was heated from –10 °C to 200 °C at a rate of 10 °C/min and was held at that temperature for 5 min to eliminate the thermal history. Afterwards, the sample was cooled to –10 °C at the same rate, and was held for about 5 min at this temperature, subsequently, being heated again to 200 °C at a rate of 10 °C/min. The degree of crystallinity (X_c %) was calculated from the heat of fusion using 207 J/g as the heat of fusion of 100% isotactic PP.

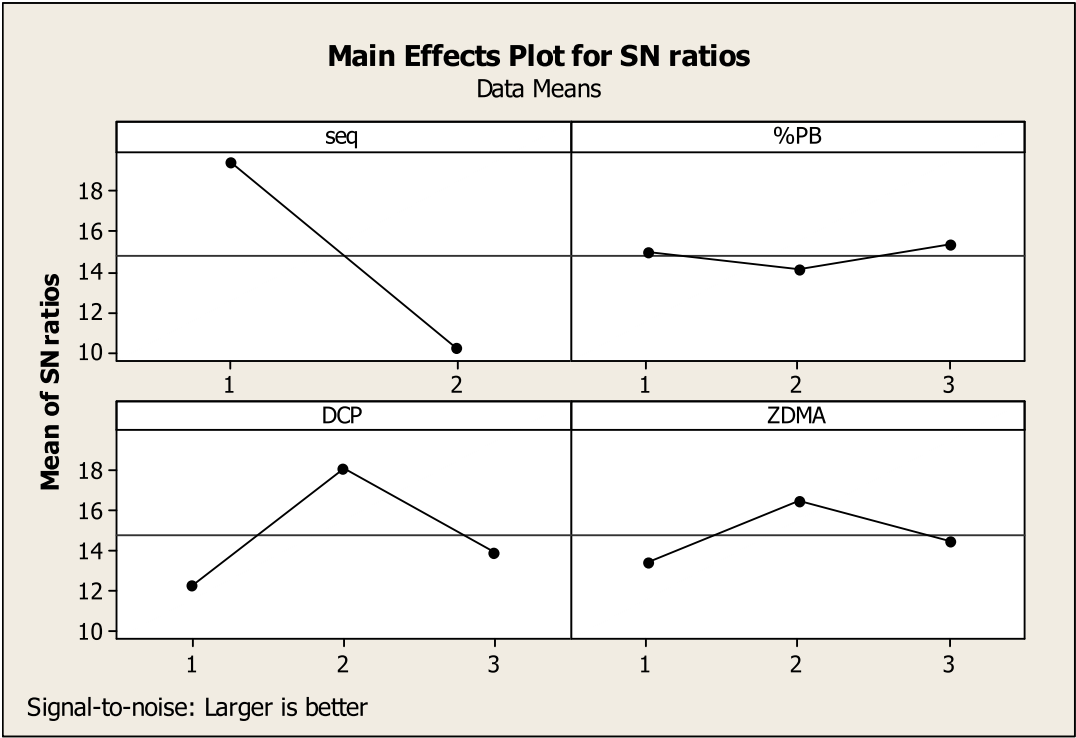
The microscopic morphology of neat PP and blends was

Table 2
Response table for means.

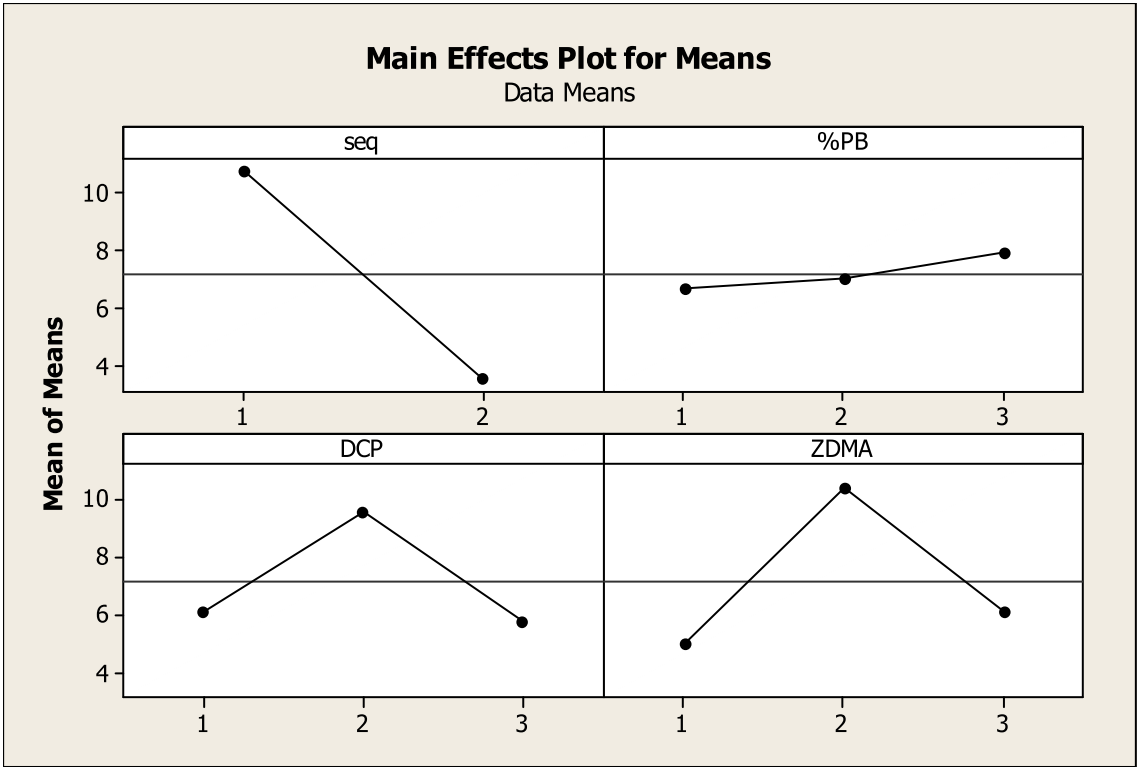
Level	DCP seq.	ZDMA (phr)	DCP (phr)	PB (Wt. %)
1	10.755	7.179	6.129	6.673
2	3.675	1.244	9.985	6.979
3	–	3.810	5.779	7.881
Δ	7.179	5.349	3.810	1.244
Rank	1	2	3	4

Table 1
Design of experiments according to L₁₈.

Sample number	Sequence of mixing	PB (Wt. %)	DCP (phr)	ZDMA (phr)	Sample number	Sequence of mixing	PB (Wt. %)	DCP (phr)	ZDMA (phr)
1	1	10	1	0	10	2	10	1	30
2	1	10	1.5	15	11	2	10	1.5	0
3	1	10	2	30	12	2	10	2	15
4	1	15	1	0	13	2	15	1	15
5	1	15	1.5	15	14	2	15	1.5	30
6	1	15	2	30	15	2	15	2	0
7	1	20	1	15	16	2	20	1	30
8	1	20	1.5	30	17	2	20	1.5	0
9	1	20	2	0	18	2	20	2	15
Reference Samples									
PP	–	–	–	–	–	–	–	–	–
A	–	10	–	–	–	–	–	–	–
B	–	15	–	–	–	–	–	–	–
C	–	20	–	–	–	–	–	–	–



A



B

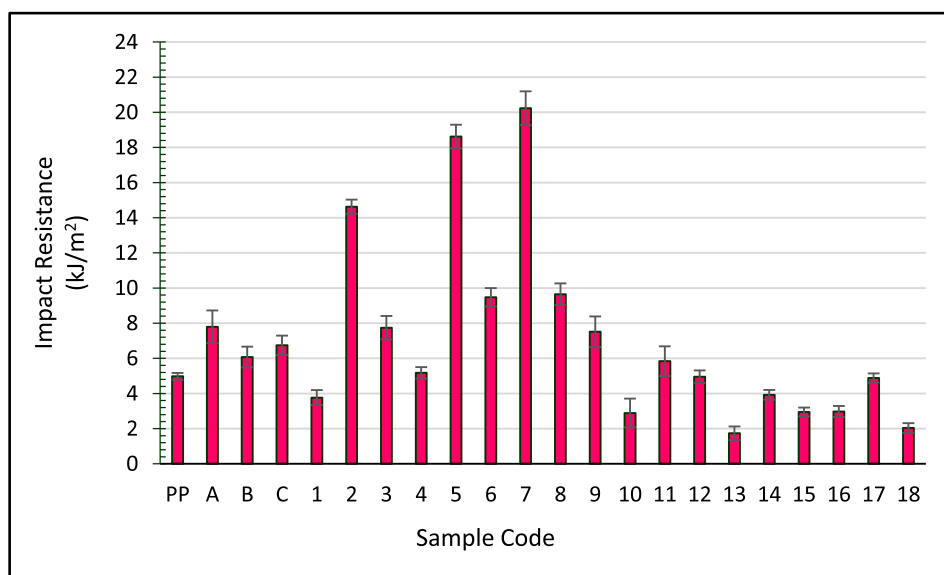
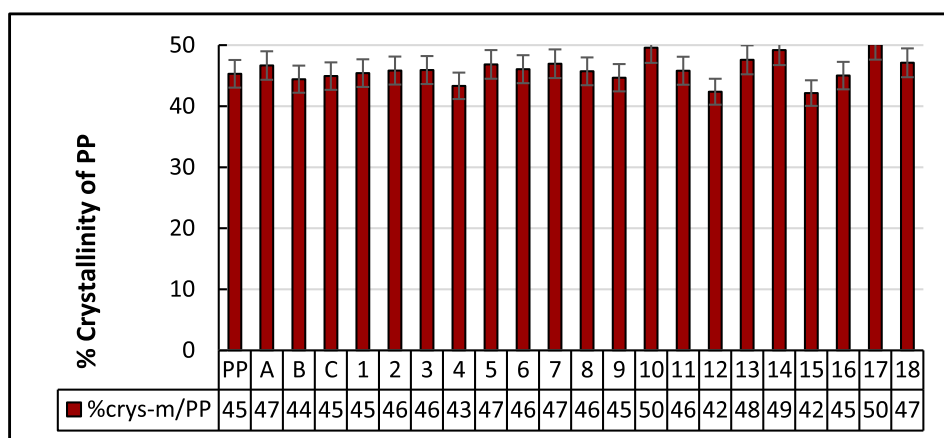
Fig. 1. Taguchi graphs of (A) S/N ratio of Impact results, (B) mean value of impact results test.

observed with a Leica DMRX POM. Samples of thin films were sandwiched between microscope cover glasses and heated to

Table 3

ANOVA analysis result for impact strength.

Parameter	Degree of freedom	Sum of square	Mean square	F	F _{critical} (%99)	%Distribution (P)
Seq. of adding DCP	1	695.850	695.850	78.556	7.3141	43.689
PB (wt. %)	(2)	14.873	Merged			
DCP (phr)	2	159.662	79.831	9.012	5.1785	9.026
ZDMA (phr)	2	261.826	145.913	16.472	5.1785	17.431
Error	48	410.326	8.858	—	—	27.034
Summation	53	1572.537	—	—	—	—

**Fig. 2.** Impact Strength of PP and its blends with PB.**Fig. 3.** Degree of crystallinity of PP and its blends with PB.

210 °C at rate of 10 °C/min with residence time of 5 min. Subsequently, the samples were cooled to room temperature.

FTIR spectroscopy analysis was performed using the KBr pellet method. Thin films of the selected samples were prepared and tested at 400–4000 cm⁻¹.

To estimate the sizes of dispersed-phase and the morphology of the blends, the impact fractured surfaces of different samples were investigated by EM3200 SEM. Before testing, all the specimens were gold-sputtered to provide enhanced conductivity.

The degree of crosslinking in the rubber and PP linked to the

rubber was measured after extraction in boiling xylene at 130 °C. The samples were dried at room temperature, after xylene solution was evaporated and subsequently weighed. The percentage of gel content of the blends was calculated using Eq. (1):

$$\text{Gel content}(\%) = (W_g/W_o) \times 100 \quad (1)$$

where W_o and W_g are sample weights before and after solvent extraction, respectively.

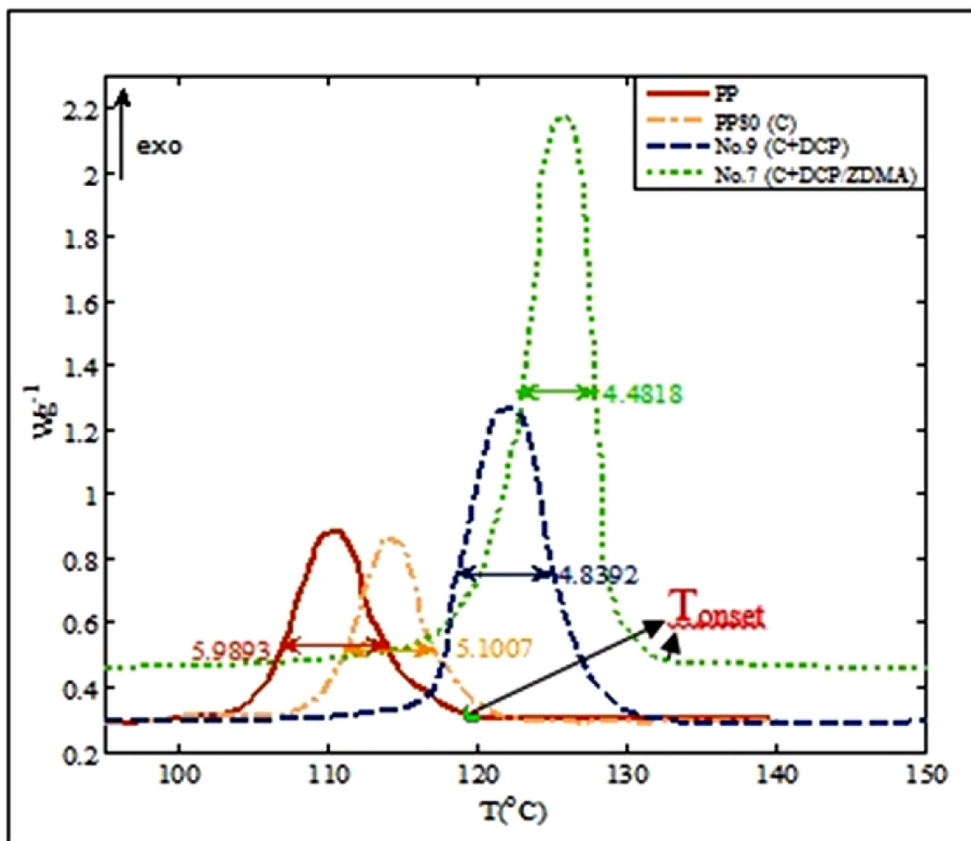


Fig. 4. Crystallization curves of PP and PP/rubber blends containing 20% PB and curing agents.

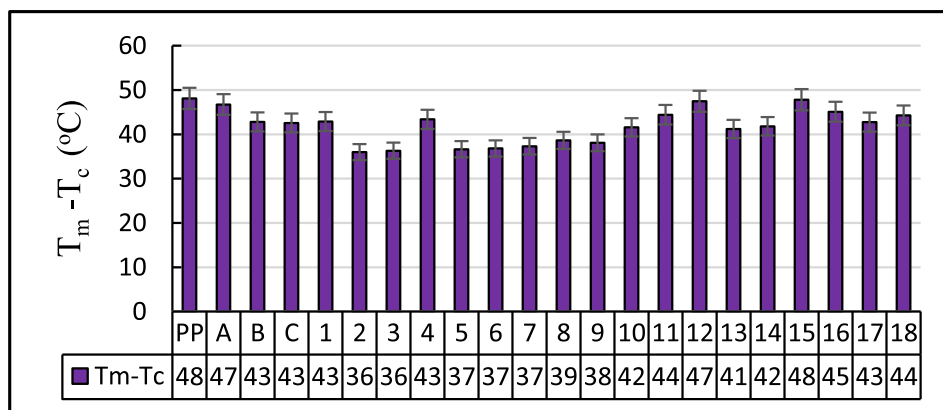


Fig. 5. Degree of supercooling for all samples according to Table 1.

3. Results and discussion

3.1. Toughness

Impact strength of the PP/PB blends was selected as the response in Taguchi analysis. The purpose of this analysis is to investigate which factor significantly affects the final product and what is the optimum experimental combination for the highest response variable. All statistical analyses were conducted by Minitab software. Table 2 shows the response table for mean values. To prepare rows of the table, the mean value of all experiments that contains level of each parameter was calculated. Then, Δ is:

$$\Delta_{\text{Factor}} = \text{response}_{\text{max}} - \text{response}_{\text{min}} \quad (2)$$

The value of Δ reflects the importance rank of each factor [28]. It is clear that the order of importance is sequence of mixing, amount of ZDMA (phr), amount of DCP (phr) and PB (wt. %).

Also, the importance of each factor was studied by signal to noise (S/N) ratio (suggested by Taguchi). S/N ratio considers both mean and variance at the same time. Larger S/N ratio reflects more relative effect of this factor. Fig. 1 (A and B) shows the main effect and S/N ratio of each factor. It is obvious that the impact strength significantly decreased by adding DCP to the rubber at first (seq. 2); this may be due to consumption of all amount of DCP in the rubber

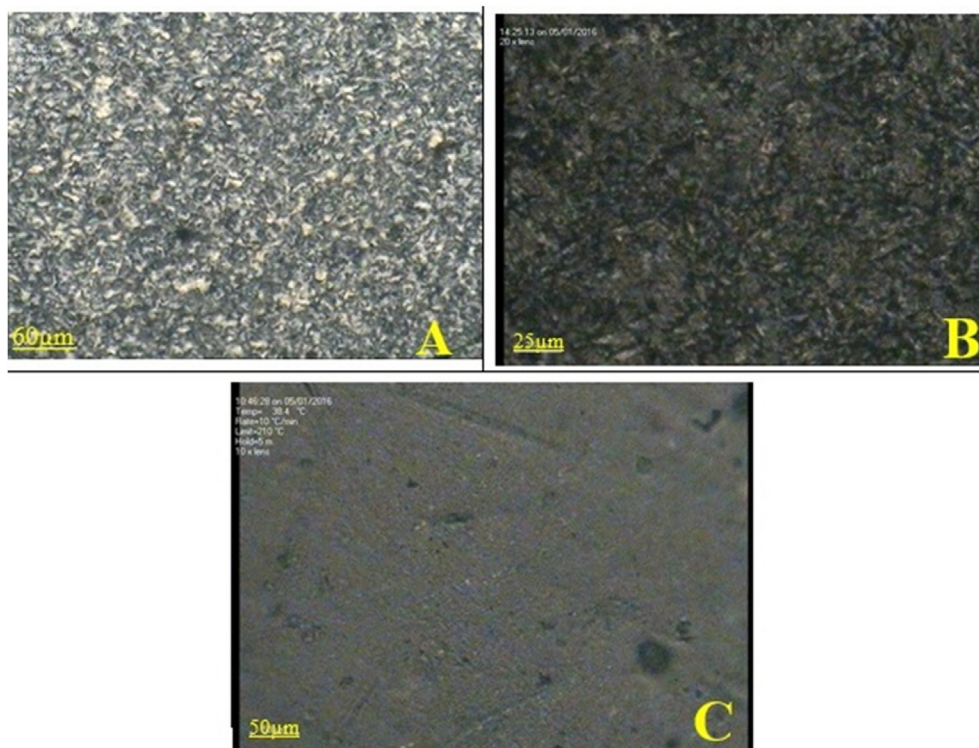


Fig. 6. POM images for (A) PP, (B) PP/PB (20% PB) without curing agent and (C) sample No.7 with curing agent.

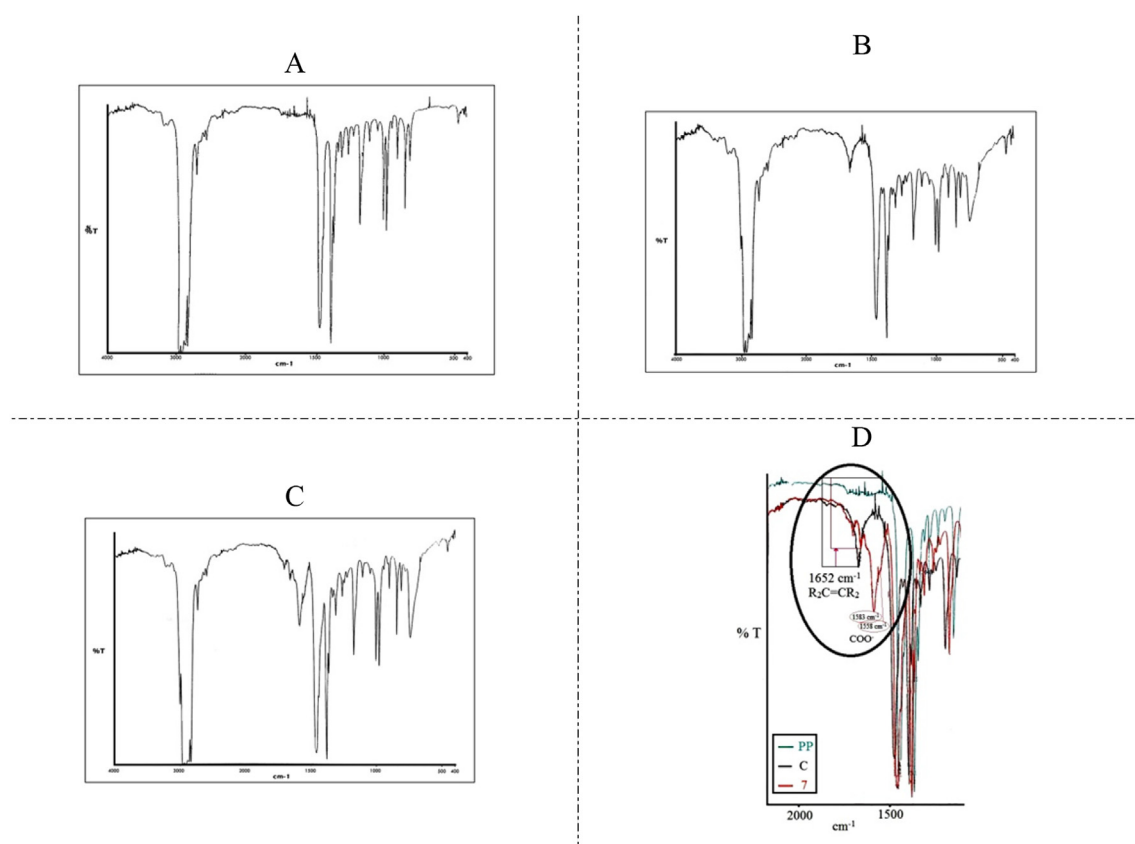


Fig. 7. FTIR spectra of (A) PP, (B) PP/PB (20% PB without curing agent), (C) sample No.7 (20% PB with curing agent) and (D) a combination of these three samples.

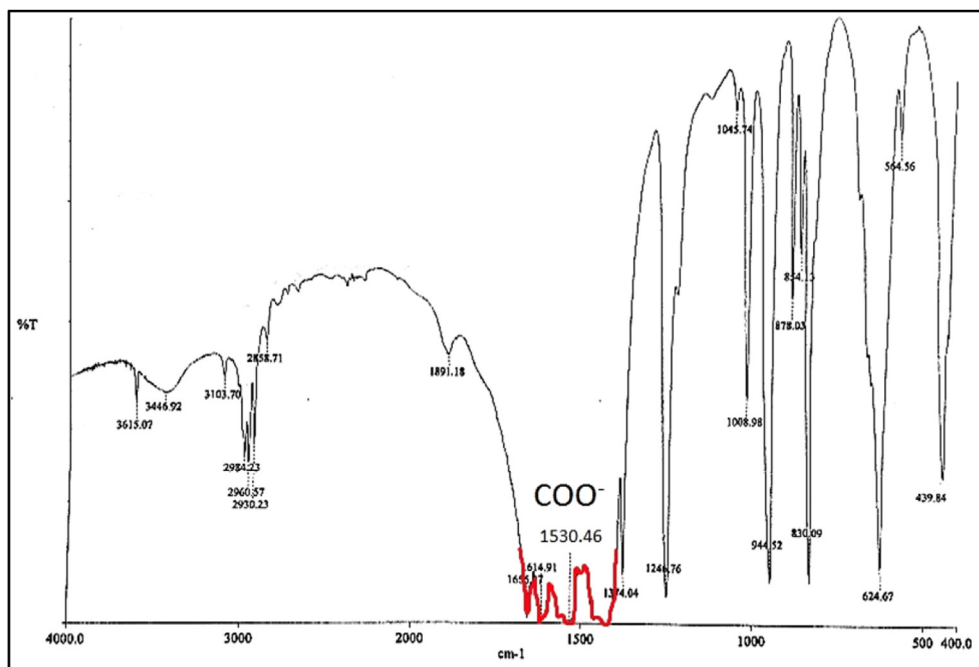


Fig. 8. ZDMA Powder FTIR spectrum. The shoulder red peaks replaced by two peak in Fig. 7 (red spectra) after polymerization of ZDMA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

domain in PB crosslinking before it can react with PP. According to the DCP curve, high amount of DCP reduces the impact strength. In addition to the degradation of PP, this may be due to the large extent of crosslinking reactions of PB and changing into rigid rubber particles [22,37].

Table 3 lists the results of ANOVA (Analysis Of Variance) which is complementary to the Taguchi analysis. The F-value gives the values of test statistic and the P-values are indicating contribution of each factor in the response.

According to Fig. 2, impact strength increased 4 times with adding 20% PB, 1phr DCP and 15phr ZDMA in seq. 1 (Sample No. 7).

The interfacial adhesion and physical mechanical properties of blends can be enhanced by in-situ compatibilization using a peroxide as well as a co-agent [12]. While crosslinking rubber with peroxide only results in the formation of a covalent bond that is quite rigid and stable, the metallic co-agent such as ZDMA-peroxide crosslink bond is “ionic” and more flexible [30]. According to Fig. 2, samples with ZDMA are tougher than the others. On the other hand, methacrylate groups in structure of ZDMA controlled the chain scission of PP by stabilizing the macro-radicals via resonance [31]. In sample No. 7, free radical reactions initiated by peroxides lead to formation of compatibilizing copolymer. This compatibilizer is a product of reaction between macro-radicals from PP chain scission and PB [32]. Also, in the presence of peroxides, the unsaturated bonds of ZDMA are opened to polymerize, and thus ZDMA can be easily grafted onto the rubber chains during peroxide crosslinking. In-situ compatibilizing of ZDMA occurs at the interfacial layer between PP and rubber through the reaction between double bonds of ZDMA and free radicals generated in both rubber and PP chains. According to the remarkable reinforcing effect of ZDMA on rubber and the possible reactions between ZDMA, rubber and PP in the free radical reaction, it is feasible to prepare PP/PB/ZDMA ternary blends via peroxide dynamic crosslinking. The above-mentioned reactions will be tracked by FTIR in the next section.

In blends with higher amount of the ZDMA, the impact strength is less than those with lower extent of the ZDMA, e.g. sample No. 8

compared to sample No. 7. More ZDMA causes ZDMA active radicals to guide the reaction towards more peroxide consumption, and only ZDMA polymerization and/or crosslinking reaction in the rubber phase will occur. This means less cross reactions as a result lower amount of compatibilizer, which reflects in impact results.

In samples No. 10 to 18, adding DCP to the rubber at first triggered rapid crosslinking in the rubber phase without any significant reactions with PP. Therefore, the mechanical properties of these samples were poor.

3.2. Crystallinity

The crystallinity of the blends were evaluated with regards to the following five parameters [26,33]: the degree of crystallization (X_c), slope of initial portion of the exotherm, which is a measure of the nucleation, crystallization inception temperature (T_{onset}), width at half-height of the peak ($w_{1/2}$), degree of supercooling ($T_m - T_c$).

Degree of crystallinity was in the range of 40–50%, as shown in Fig. 3, and not affected appreciably by the presence of PB in the blends.

Fig. 4 compares sample No.7 with the neat PP as well as its counterparts, i.e. blends with similar PP/PB composition. Accordingly, widths of the crystallization peak at half height ($w_{1/2}$) are smaller and slopes of these curves are greater than that of the neat PP, which means addition of curing agents (DCP and/or ZDMA) increases the crystallization rate of the blends. This implies PB and curing system, especially ZDMA, act as nucleating agents in the crystallization of PP.

Decrease in degree of supercooling of the sample No.7 is evident from Fig. 5. This can be related to the nucleation effect of PB and ZDMA in accord with increase in T_{onset} in Fig. 4 [17]. The role of ZDMA and PB as nucleating agents has two main consequences: 1) increase of crystallization rate; 2) decrease in size of spherulites, which is of great importance in improving the impact strength of PP. It is generally accepted that impact resistance is inversely related to spherulite size [34]. This may be due to decrease of strained amorphous ties between spherulites, or defects within the

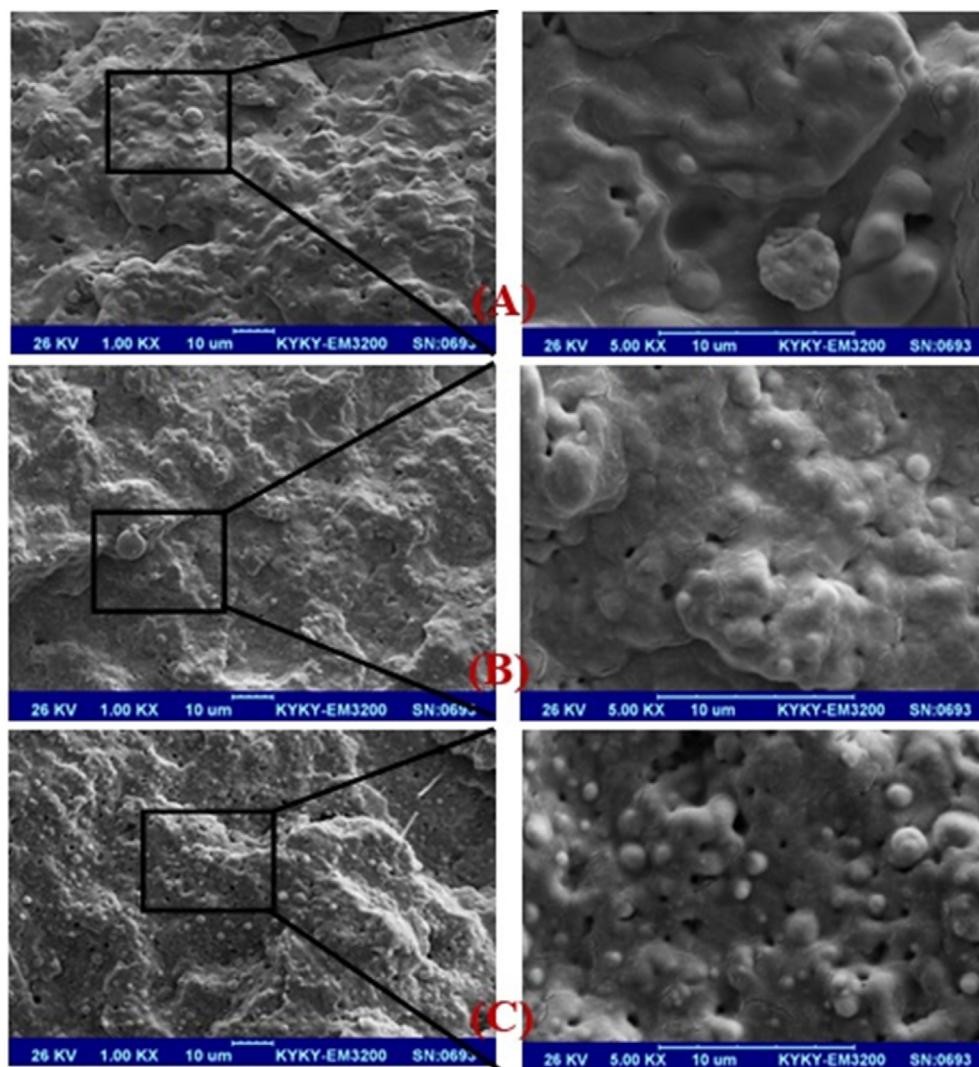


Fig. 9. SEM images of the (A) PP/PB blend (B) sample No.9 (curing without ZDMA), (C) sample No.7. All the samples have 20% PB.

spherulites themselves which are formed during the growth step. These stress concentration zones will be distributed uniformly in the matrix when spherulites become smaller leading to prevention of catastrophic failure under stress and increase of impact resistance [34].

In Fig. 6, crystal structure of PP and blends is shown. In the samples with highest amount of rubber, both with and without curing agent, size of spherulites is reduced and their boundaries are not sharp (No.7 and C). This agrees with increase of impact strength of this sample.

3.3. Reactions tracking

To follow reactions during compatibilization, FTIR spectra of the neat PP and blends of PP/PB with 20 wt. % rubber were acquired, as shown in Fig. 7. The bands at 1652 cm^{-1} are characteristic of $\text{R}_2\text{C}=\text{CR}_2$, which are assigned to the stretching mode of $\text{C}=\text{C}$. They can be used to detect the ZDMA content and double bonds of rubber. In sample No.7 with in situ compatibilization, the intensity of 1652 cm^{-1} bands decreased during the cure, implying the consumption of double bonds and the occurrence of reaction between ZDMA and PB during peroxide curing.

The spectrum of the zincdimethacrylate is shown in Fig. 8. The

carboxylate anion stretching vibrations of $\text{Zn}(\text{MA})_2$ are represented by several bands in the $1600\text{--}1400\text{ cm}^{-1}$ region. After the polymerization, these bands are replaced by two bands centered at 1599 and 1422 cm^{-1} . This shows that the in situ radical polymerization of ZDMA takes place during the curing process with peroxide [35]. As mentioned in the previous section, during this process, PP/PB/poly-ZDMA ternary blend is produced and in situ compatibilization occurred.

3.4. Morphology

Parameters such as shape, size and size distribution of the dispersed rubber particles are among important factors affecting the toughening role of the rubbery phase. At similar rubber content, PP blends with smaller rubber particles (about $0.5\text{--}1\text{ }\mu\text{m}$) are tougher and more ductile than those with larger particles [36,37]. This is attributed to the more efficient action of the smaller rubber particles in induction of crazing and/or shear yielding to the matrix [3]. According to SEM micrographs, addition of ZDMA lead to size reduction and more uniform size distribution of the crosslinked PB phase, as shown in Fig. 9. Particle size reduction can be assigned to reduction of interfacial tension. In Fig. 9(A) and (B), aggregations can be seen in samples without ZDMA, especially in the

Table 4
Gel content of blends with 20% PB.

Sample code no.	DCP amount (phr)	ZDMA amount (phr)	Gel content (%)
7	1	15	74.66
8	1.5	30	66.13
9	2	0	65.64

uncrosslinked samples. These are prone to crack formation and premature failure. Consequently, ZDMA not only produces strong interaction between the two phases [18], but also helps to disperse and distribute the rubber particles.

Interestingly, in SEM micrograph of sample No. 7, one can see some cracks originated uniformly from each rubber particle. This may be described according to the Wu's criterion. Wu has reported a sharp brittle-ductile transition of a polymer/rubber binary blend at a critical surface-to-surface inter-particle distance:

$$L_c = d_c \left[k(\pi/6\phi_r)^{1/3} - 1 \right] \quad (3)$$

where d_c is the critical rubber particle diameter, k is the geometric constant (for example, $k = 1$ for cubic lattice), ϕ_r is the rubber volume fraction. L_c is the average surface to surface inter-particle distance, and depends on the rubber volume fraction. If $L < L_c$, continuum percolation of stress volume around rubber particles will occur and the matrix yielding will propagate and pervade over the entire matrix, resulting in good toughness of the blend. The average rubber particle diameters for sample No.7 obtained from SEM photographs is about 0.63 μm which is in the suitable range of 0.5–1 μm for PP [37].

3.5. Gel content measurements

Effect of adding curing agents to PP/PB on the gel content is shown in Table 4. The amount of gel was determined after extraction of uncrosslinked rubber phase and unreacted PP with rubber in the network. Gel is formed when a strong interaction between components of the blend exists. It demonstrates that crosslinks are

formed. As seen in Table 4, in sample No.7 with the maximum impact strength, the percentage of gel is the highest. This is supported by torque curves during compounding. As shown in Fig. 10, the steady-state torque of sample No. 7 is higher than its counterparts (samples with the same amount of rubber which were mixed similarly), indicating greater viscosity as a result of higher crosslink density, as well as interactions between components.

The more the amount of DCP, the lower the gel content. This might be due to PP chain scission. However, the gel content was not changed according to the amount of ZDMA. In other words, although, ZDMA prevented PP degradation in sample No. 8, it also consumed DCP in its polymerization.

It is worth noting that, for all curves in Fig. 10, the two large peaks are related to addition of PP and PB. A drop after adding DCP in the green curve indicates the degradation of PP matrix that has been moderated by adding ZDMA in the black curve. Incorporation of ZDMA results in increase of the final torque. This may be also due to the reinforcing effect of ZDMA in addition to its compatibilizing role [18].

4. Conclusions

According to this study, in the presence of an adequate amount of curing agents, polybutadiene rubber can be used to improve the toughness of polypropylene. Results demonstrated that the blend with 20 wt. % rubber, 1phr DCP, 15phr ZDMA that was compounded using the first procedure of mixing, has an impact strength almost four times greater than neat PP. Taguchi analysis shows that the effect of sequence of DCP addition and the amount of ZDMA are the most important factors affecting the blends properties. Increase in the toughness is thought to be result of the nucleation effect of rubber and co-agent particles that reduces the size and size distribution of the PP spherulites, and makes a finer and more uniform crystalline phase, as well as reduction of rubber particle size and their better dispersion due to applying the curing system. In this regard, as evident from FTIR data, in situ compatibilization plays a crucial role both in controlling size and distribution of the PB phase and, also, its adhesion to the PP matrix which is important from stress transfer point of view. In the second sequence of mixing, rapid crosslinking of the rubber phase affected reactivity with PP.

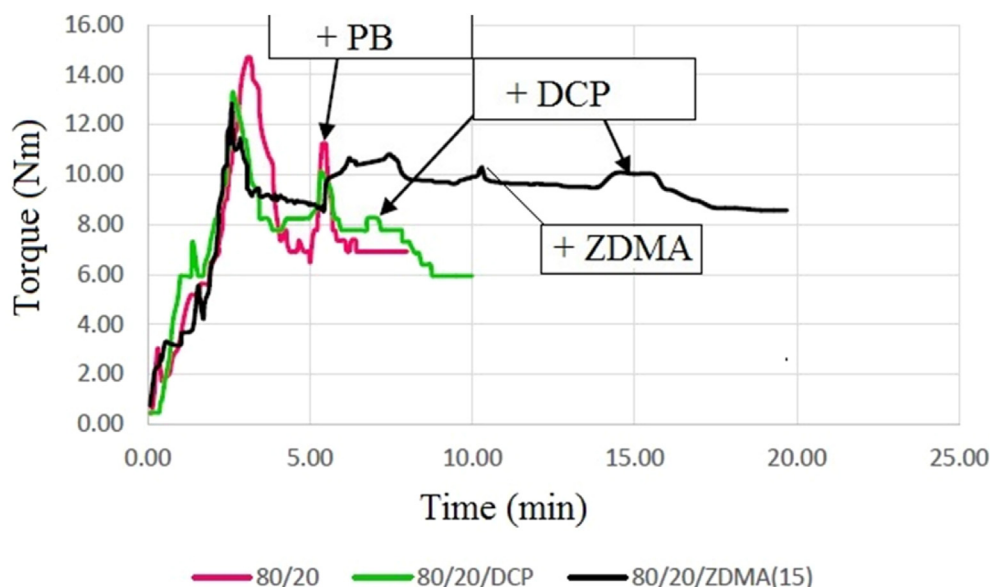


Fig. 10. Torque curves during processing for PP/PB blends with 20 wt. % rubber and different amount of curing agents.

This results in rigid rubber particles and prevents sufficient copolymer formation and, therefore, an incompatible blend will be produced.

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